

Name: Raghbir Rizwan Rabani
NU ID: 23K-0012
Instructor: Sir Omar Qureshi

ASSIGNMENT # 1

Reflective Summary

Introduction and Reflection:

The lecture on Variational Autoencoders (VAEs) helped me understand how an ordinary Autoencoder can be extended to become a generative model. A standard Autoencoder compresses an input into a fixed latent vector and reconstructs it, but its latent space is not properly organized, so random sampling can produce unrealistic outputs. A VAE addresses this problem by learning a probability distribution in the latent space, making it continuous and suitable for generating new data.

How a VAE Works:

In a VAE, the encoder does not directly produce one latent vector. Instead, it produces a mean μ and standard deviation σ for a probability distribution $q\phi(z|x)$. A latent vector is sampled from this distribution and sent to the decoder. The sampling step is written as $z = \mu + \sigma \odot \epsilon$, where $\epsilon \sim N(0, I)$. This is called the reparameterization trick. It separates the random part of sampling from the trainable parameters, allowing gradients to pass through μ and σ during backpropagation.

Loss Functions in a VAE:

The reconstruction part measures how close the reconstructed output $\hat{x}$ is to the original input x . For continuous data, Mean Squared Error (MSE) can be used: $L_MSE = (1/n) \sum_i (x_i - \hat{x}_i)^2$. For binary or normalized data, Binary Cross-Entropy (BCE) can be used: $L_BCE = -\sum_i [x_i \log(\hat{x}_i) + (1 - x_i) \log(1 - \hat{x}_i)]$. In a VAE, the reconstruction loss can be viewed as the negative expected log-likelihood, $L_reconstruction = E_{q\phi(z|x)}[-\log p\theta(x|z)]$.

The second important component is KL divergence. It measures how different the encoder distribution $q\phi(z|x)$ is from the standard normal prior $p(z) = N(0, I)$. For a Gaussian posterior, it is calculated as $L_KL = -1/2 \sum_i [1 + \log(\sigma_i^2) - \mu_i^2 - \sigma_i^2]$. This term encourages the latent distributions to remain close to a standard normal distribution, producing a smooth and continuous latent space. Therefore, the overall VAE loss is $L_VAE = L_reconstruction + L_KL$.

Difference Between VAE and β -VAE:

The main difference between a VAE and a β -VAE is the weight given to the KL-divergence term. In a standard VAE, $\beta = 1$, so the loss is $L_VAE = L_reconstruction + L_KL$. In β -VAE, β is normally greater than 1, giving more importance to regularizing the latent space: $L_beta\text{-VAE} = L_reconstruction + \beta L_KL$. Increasing β encourages different latent dimensions to represent more independent or disentangled factors, which makes the learned representation easier to interpret. However, too much KL regularization can reduce reconstruction quality and make outputs overly smooth or blurry. Thus, VAE mainly balances reconstruction and regularization, while β -VAE places stronger emphasis on obtaining a structured and disentangled latent representation.

Conclusion:

Overall, the lecture showed me that VAEs are more than compression models because they learn a meaningful probabilistic latent space from which new samples can be generated. The most important concepts for me were the reparameterization trick, the reconstruction and KL losses, and the effect of β in β -VAE. Understanding these concepts makes it clear how VAEs can be used for representation learning, generation, and other machine learning applications.